Digital Image Processing

Image Enhancement

(Point Processing)

Introduction to the Basic Mathematical Tools

- For example, consider the following images (matrices):

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}$$

The *elementwise product* (often denoted using the symbol $\odot$ or $\otimes$) of these two images is

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \odot \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} = \begin{bmatrix} a_{11}b_{11} & a_{12}b_{12} \\ a_{21}b_{21} & a_{22}b_{22} \end{bmatrix}$$

Arithmetic Operations

- Arithmetic operations between two images f(x, y) and g(x, y) are denoted as

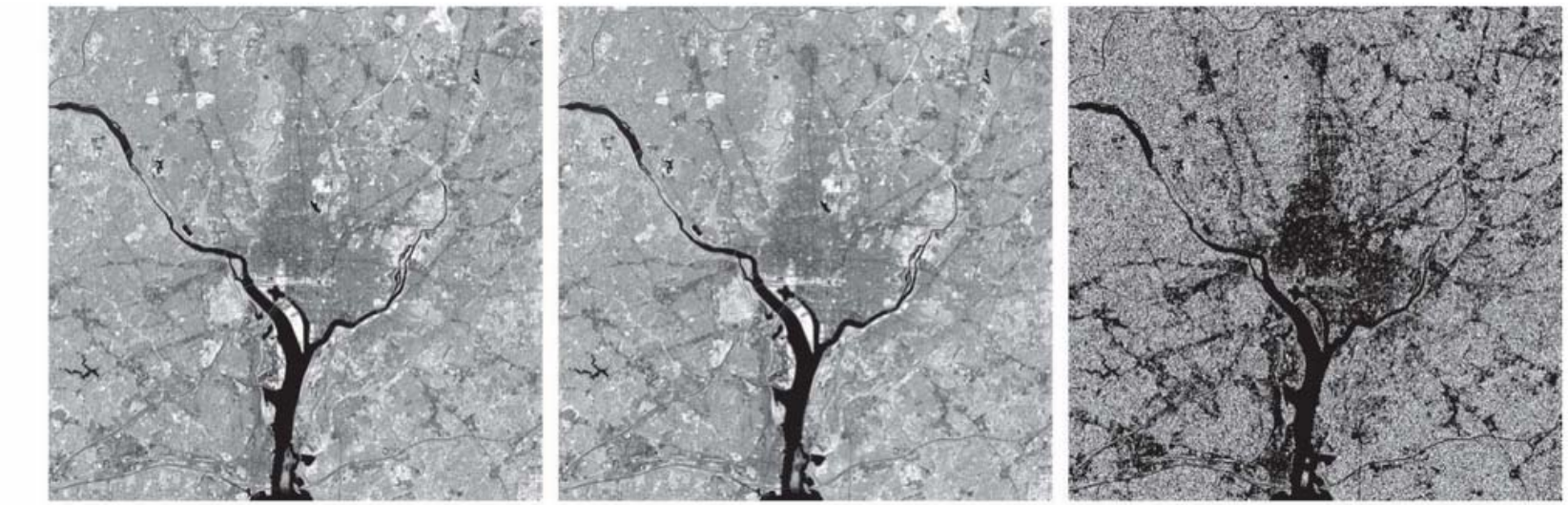
$$s(x,y) = f(x,y) + g(x,y)$$

$$d(x,y) = f(x,y) - g(x,y)$$

$$p(x,y) = f(x,y) \times g(x,y)$$

$$v(x,y) = f(x,y) \div g(x,y)$$

Image subtraction



a b c

FIGURE 2.30

(a) Infrared image of the Washington, D.C. area. (b) Image resulting from setting to zero the least significant bit of every pixel in (a). (c) Difference of the two images, scaled to the range [0, 255] for clarity.

(Original image courtesy of NASA.)

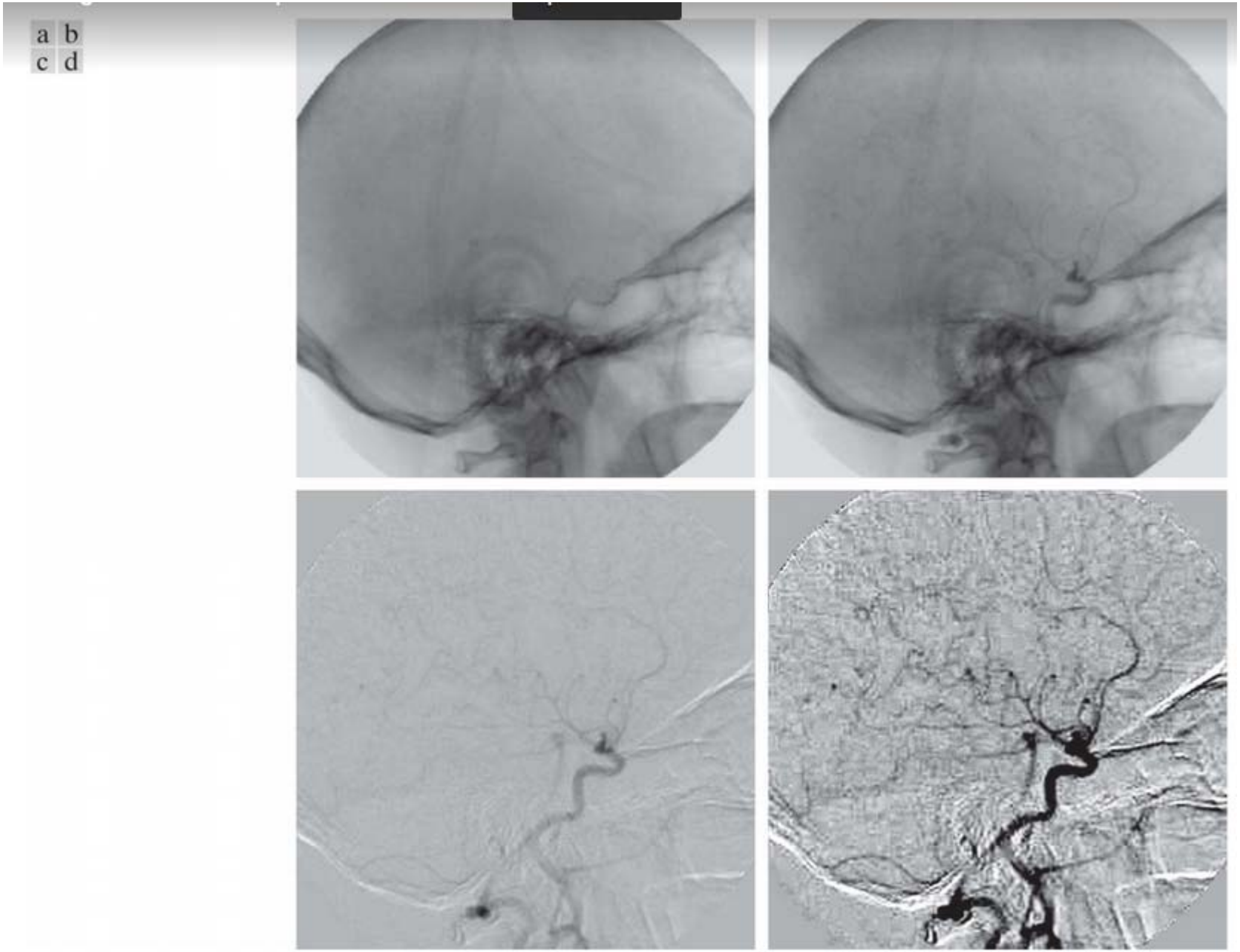


FIGURE 2.32

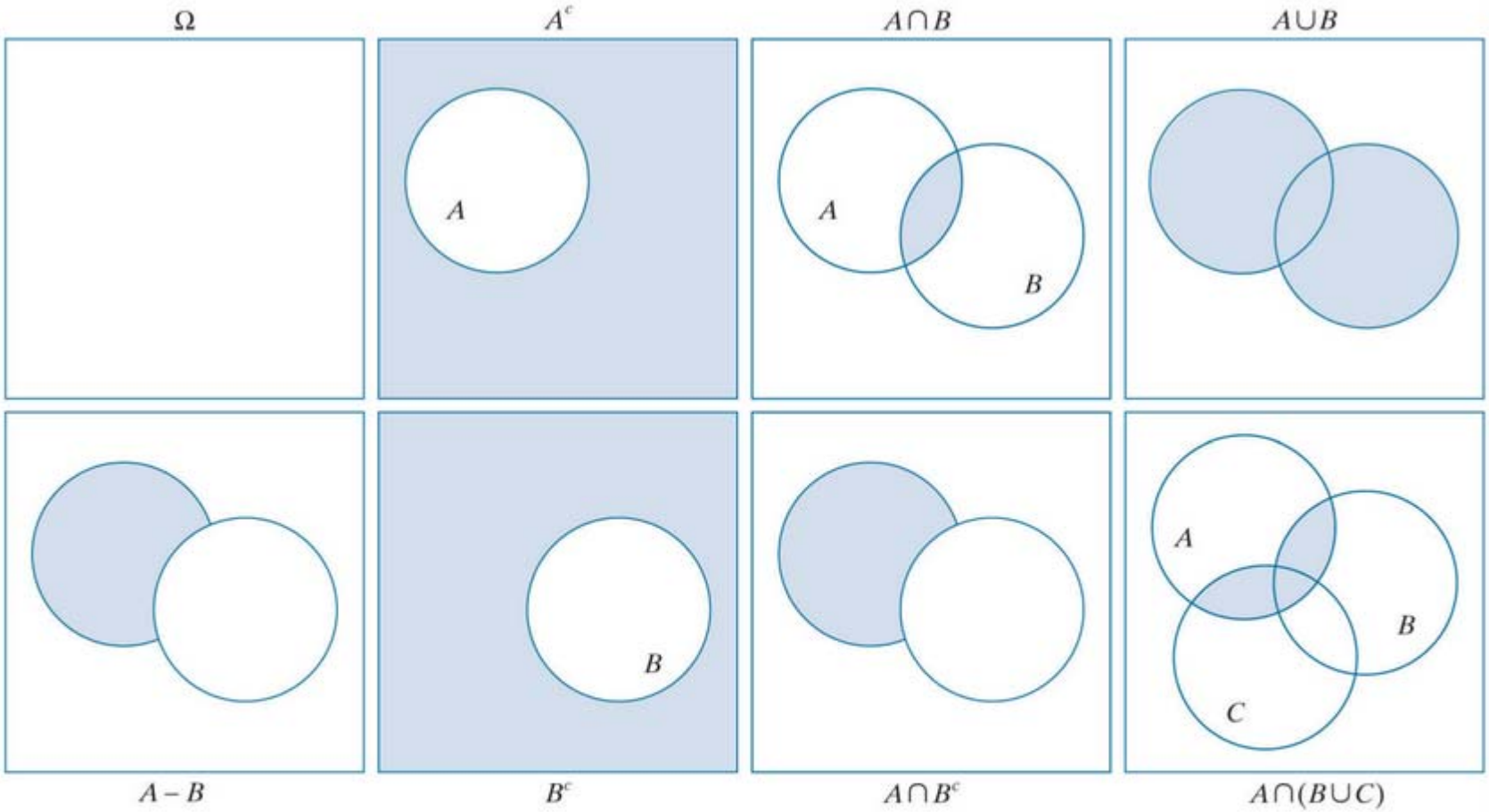
Digital subtraction angiography. (a) Mask image. (b) A live image. (c) Difference between (a) and (b). (d) Enhanced difference image.



a b c

FIGURE 2.34

(a) Digital dental X-ray image. (b) ROI mask for isolating teeth with fillings (white corresponds to 1 and black corresponds to 0). (c) Product of (a) and (b).



a b c d
e f g h

FIGURE 2.35

Venn diagrams corresponding to some of the set operations in Table 2.1 . The results of the operations, such as A^c, are shown shaded. Figures (e) and (g) are the same, proving via Venn diagrams that $A - B = A \cap B^c$ [see Eq. (2-40)].

a b c

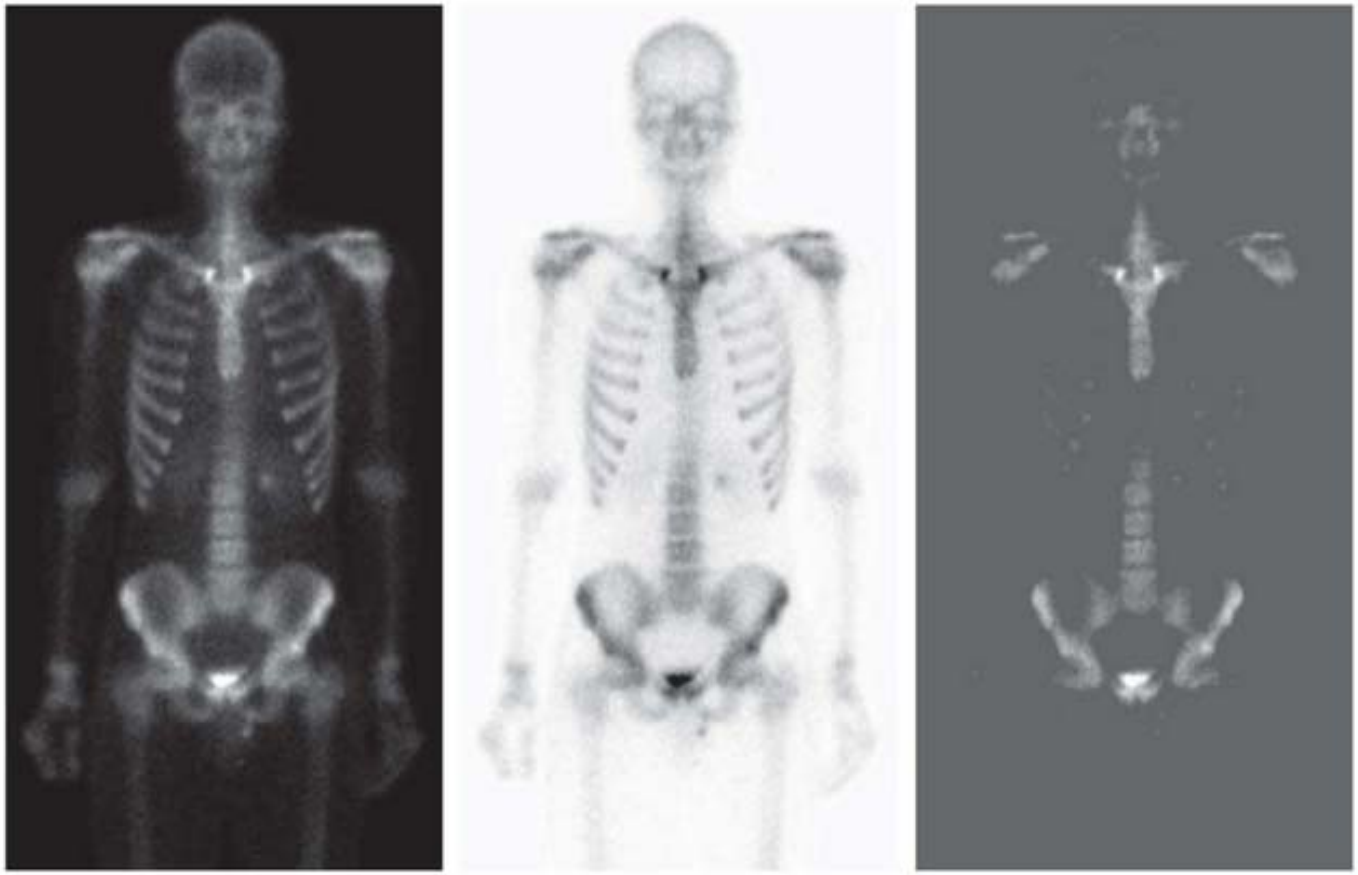


FIGURE 2.36
Set operations involving grayscale images. (a) Original image. (b) Image negative obtained using grayscale set complementation. (c) The union of image (a) and a constant image.

The *union* of two grayscale sets A and B with the same number of elements is defined as the set

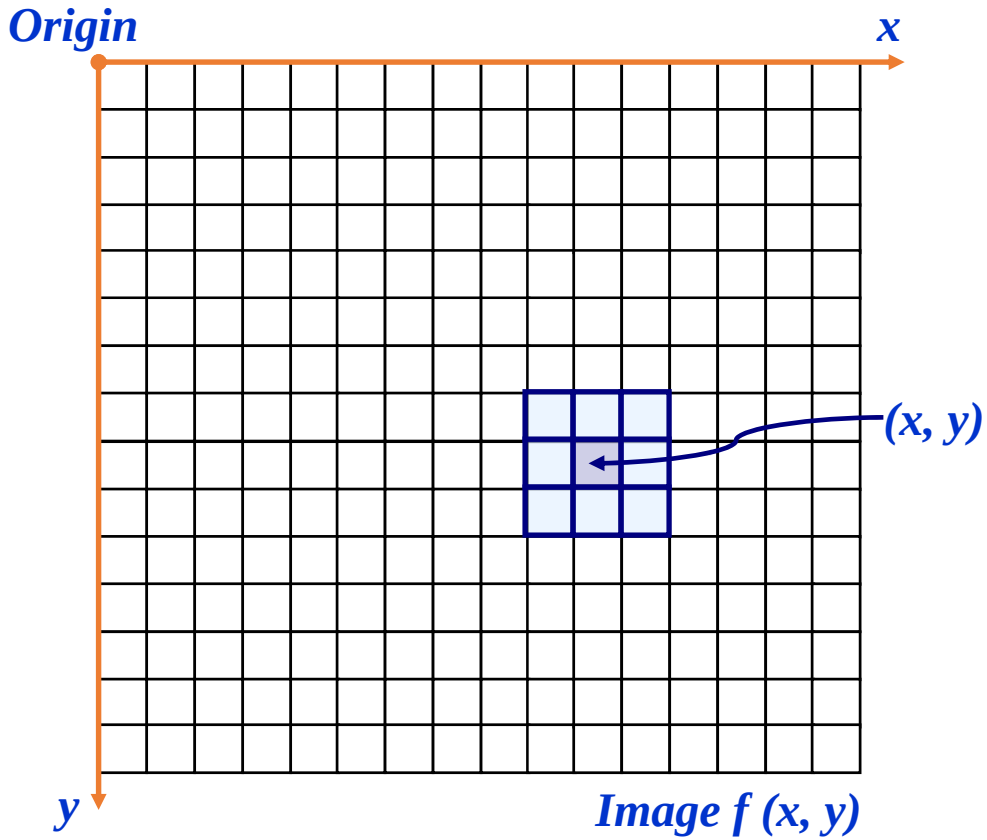
$$A \cup B = \left\{ \max_z(a, b) \mid a \in A, b \in B \right\}$$

Contents

- In this lecture we will look at image enhancement point processing techniques:
  - What is point processing?
  - Negative images
  - Thresholding
  - Logarithmic transformation
  - Power law transforms
  - Grey level slicing
  - Bit plane slicing

Basic Spatial Domain Image Enhancement

- Most spatial domain enhancement operations can be reduced to the form
- $g(x, y) = T[f(x, y)]$
- where $f(x, y)$ is the input image, $g(x, y)$ is the processed image and T is some operator defined over some neighbourhood of (x, y)



Point Processing

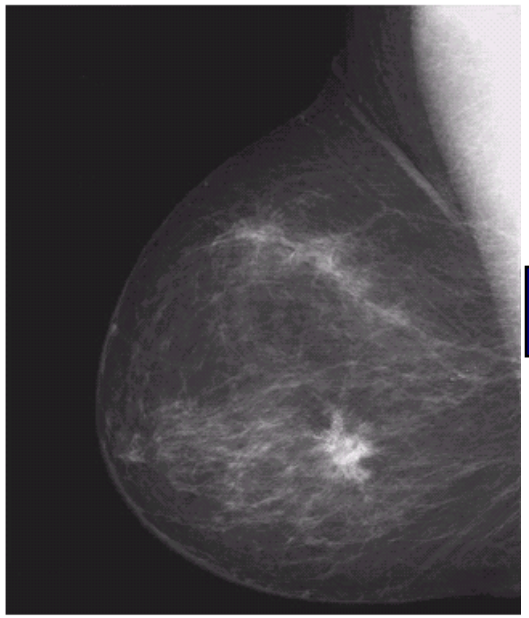
- The simplest spatial domain operations occur when the neighbourhood is simply the pixel itself
- In this case T is referred to as a *grey level transformation function* or a *point processing operation*
- Point processing operations take the form
  - $s = T(r)$
- where S refers to the processed image pixel value and r refers to the original image pixel value

Point Processing Example: Negative Images

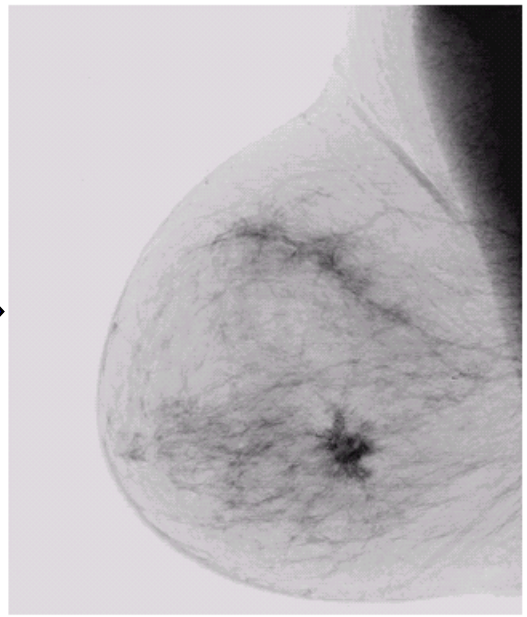
Images taken from Gonzalez & Woods, Digital Image Processing (2002)

- Negative images are useful for enhancing white or grey detail embedded in dark regions of an image
  - Note how much clearer the tissue is in the negative image of the mammogram below

Original Image

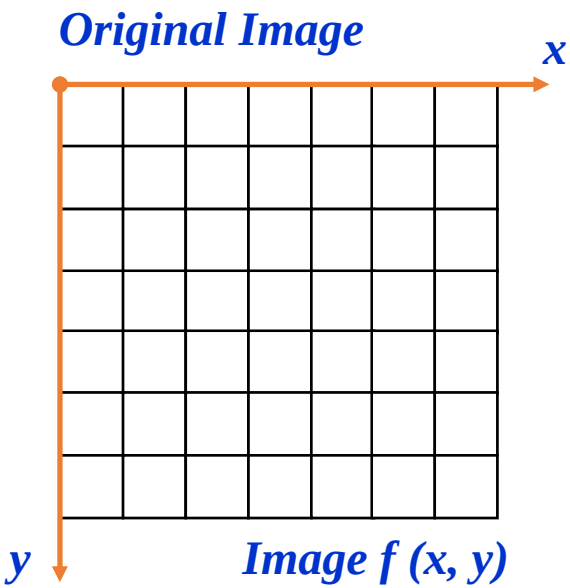


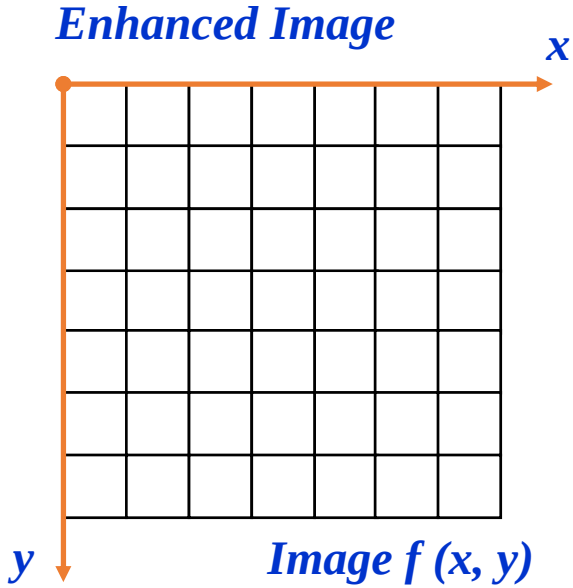
$s = 1.0 - r$



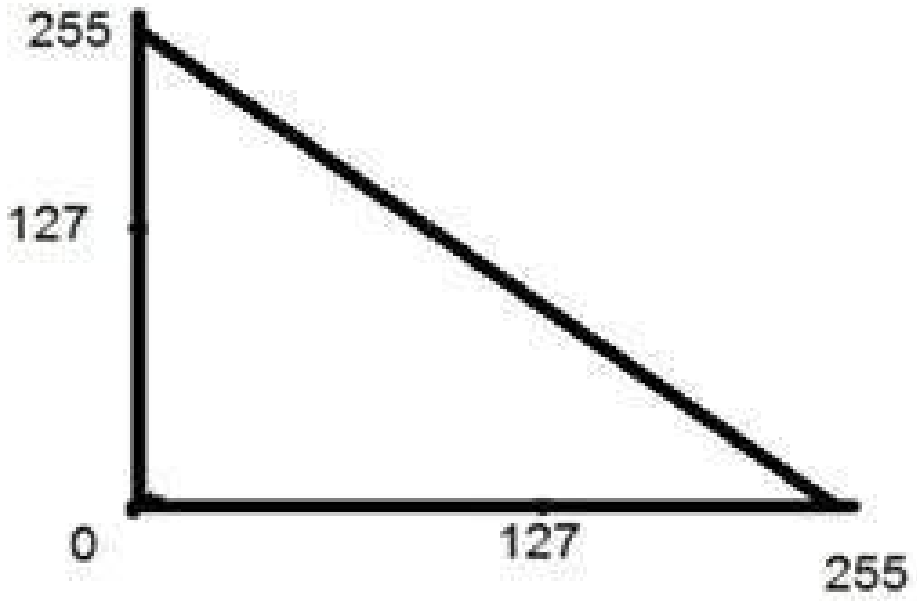
Negative Image

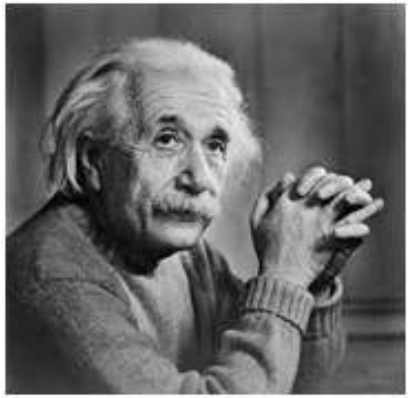
Point Processing Example: Negative Images (cont…)





$$s = intensity_{max} - r$$

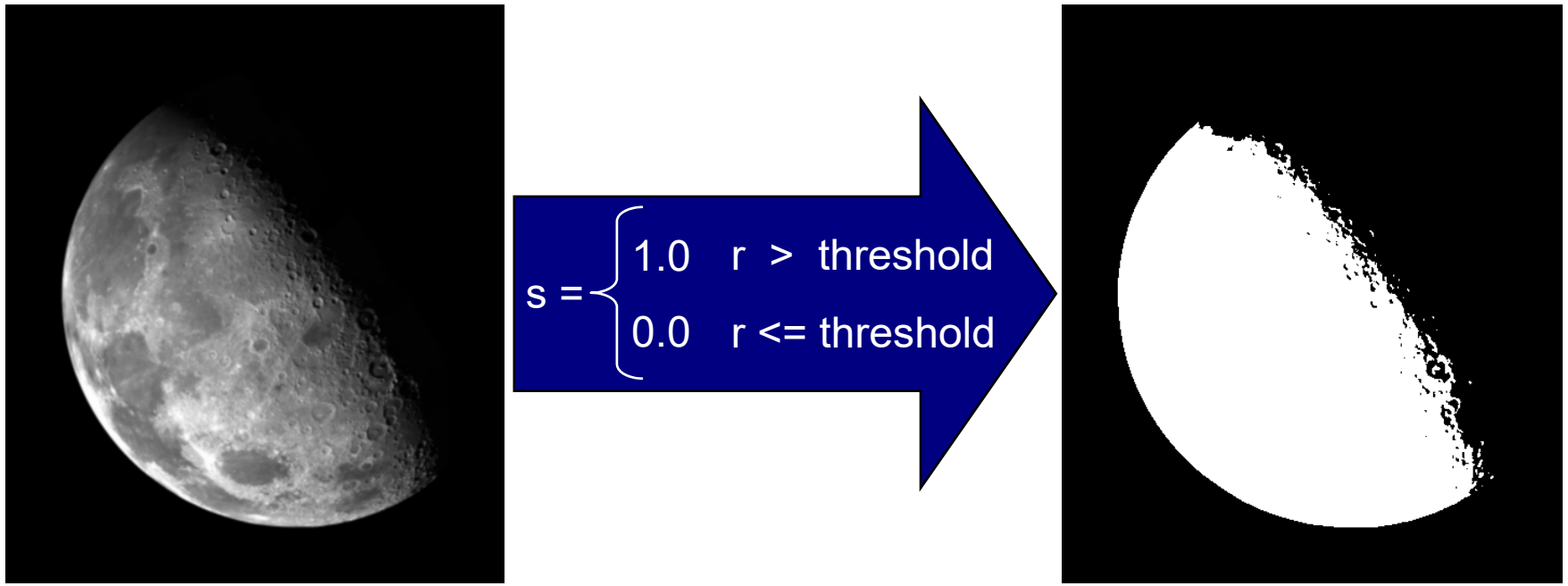

255
127
0
127
255





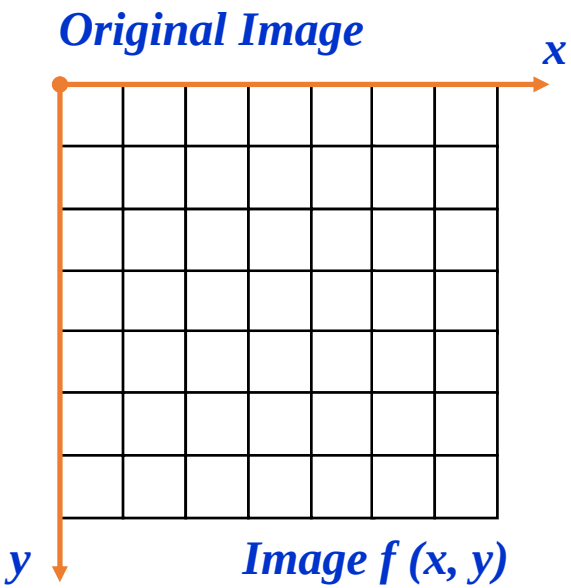
Point Processing Example: Thresholding

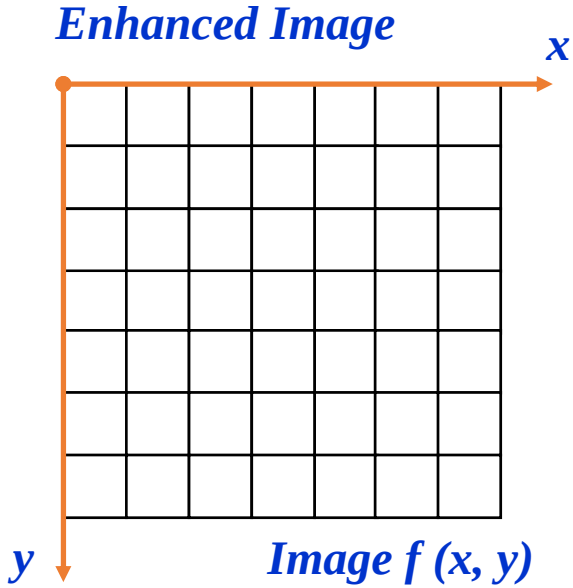
•Thresholding transformations are particularly useful for segmentation in which we want to isolate an object of interest from a background



Images taken from Gonzalez & Woods, Digital Image Processing (2002)

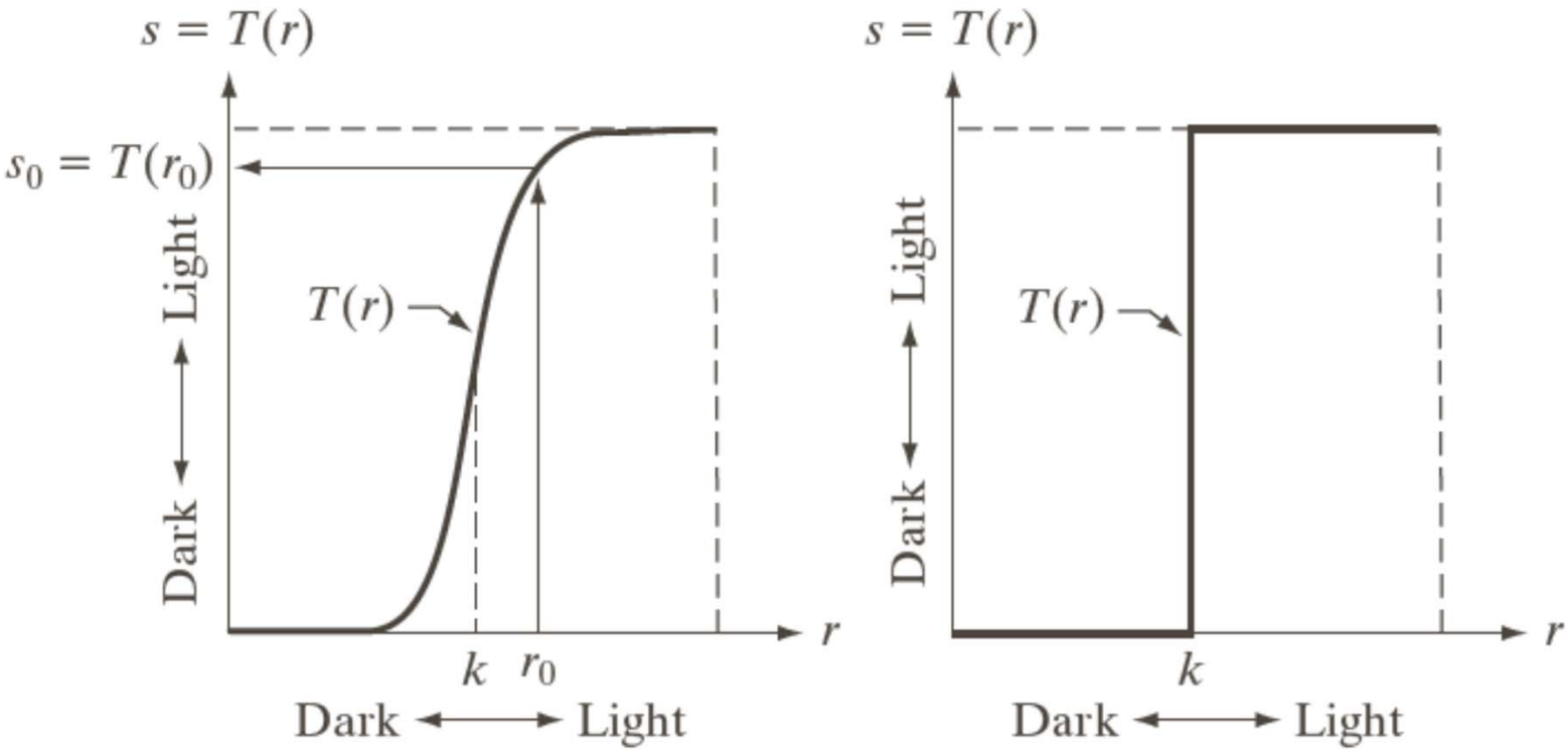
Point Processing Example: Thresholding (cont…)





$$s = \begin{cases} 1.0 & r > threshold \\ 0.0 & r <= threshold \end{cases}$$

Intensity Transformations

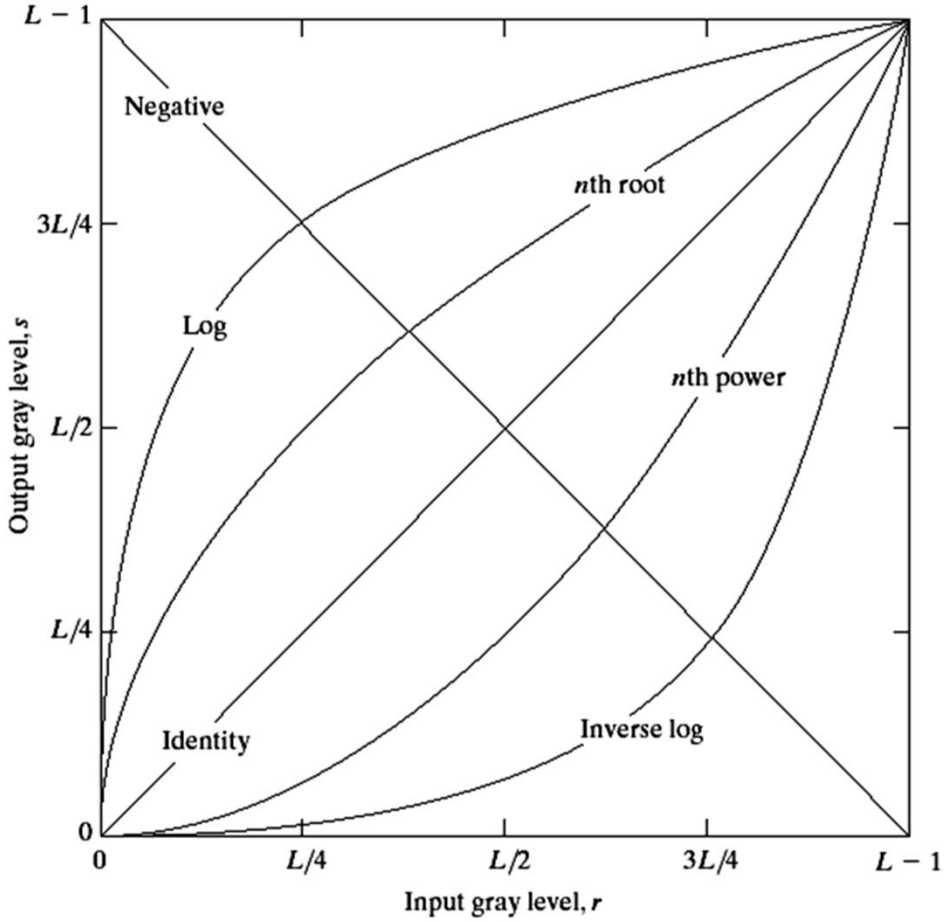


Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Basic Grey Level Transformations

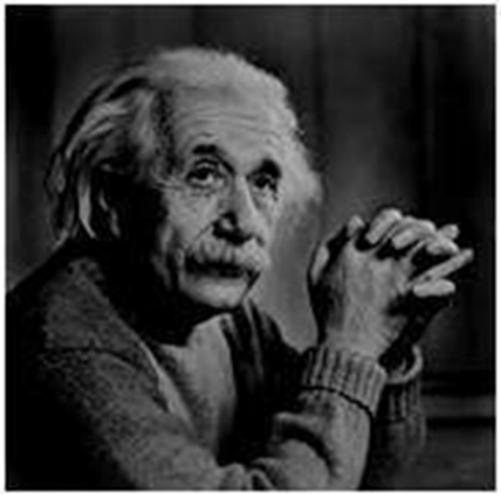
Images taken from Gonzalez & Woods, Digital Image Processing (2002)

- There are many different kinds of grey level transformations
- Three of the most common are shown here
  - Linear
    - Negative/Identity
  - Logarithmic
    - Log/Inverse log
  - Power law
    - n^{th} power/n^{th} root



Logarithmic Transformations

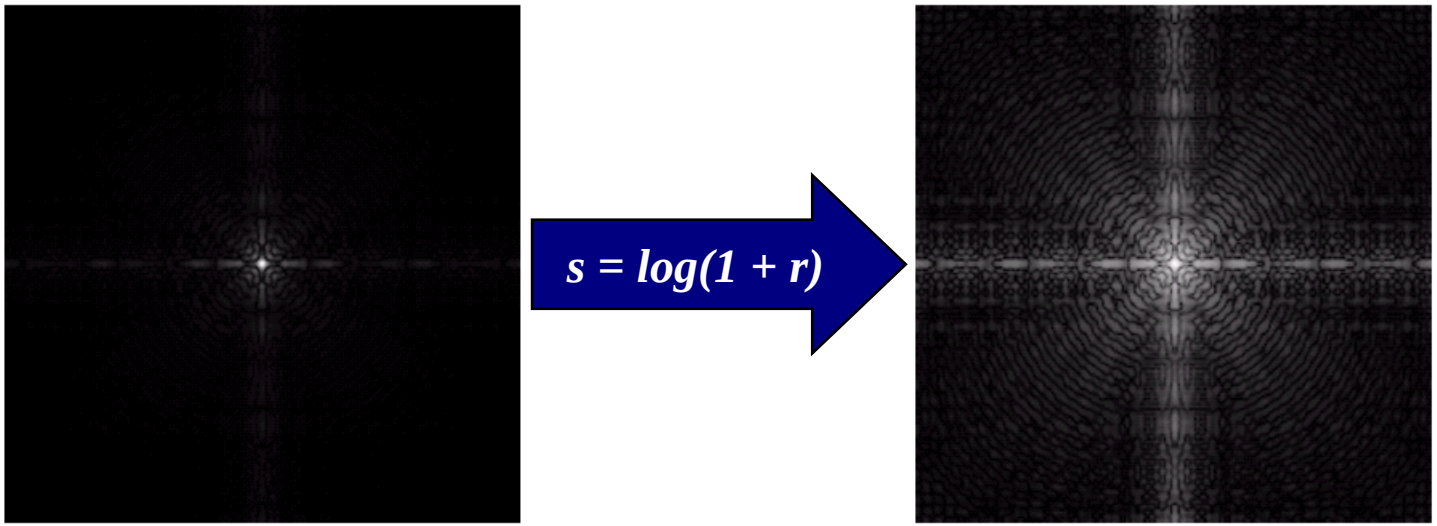
- The general form of the log transformation is
  - $s = c * log(1 + r)$
- The log transformation maps a narrow range of low input grey level values into a wider range of output values
- The inverse log transformation performs the opposite transformation





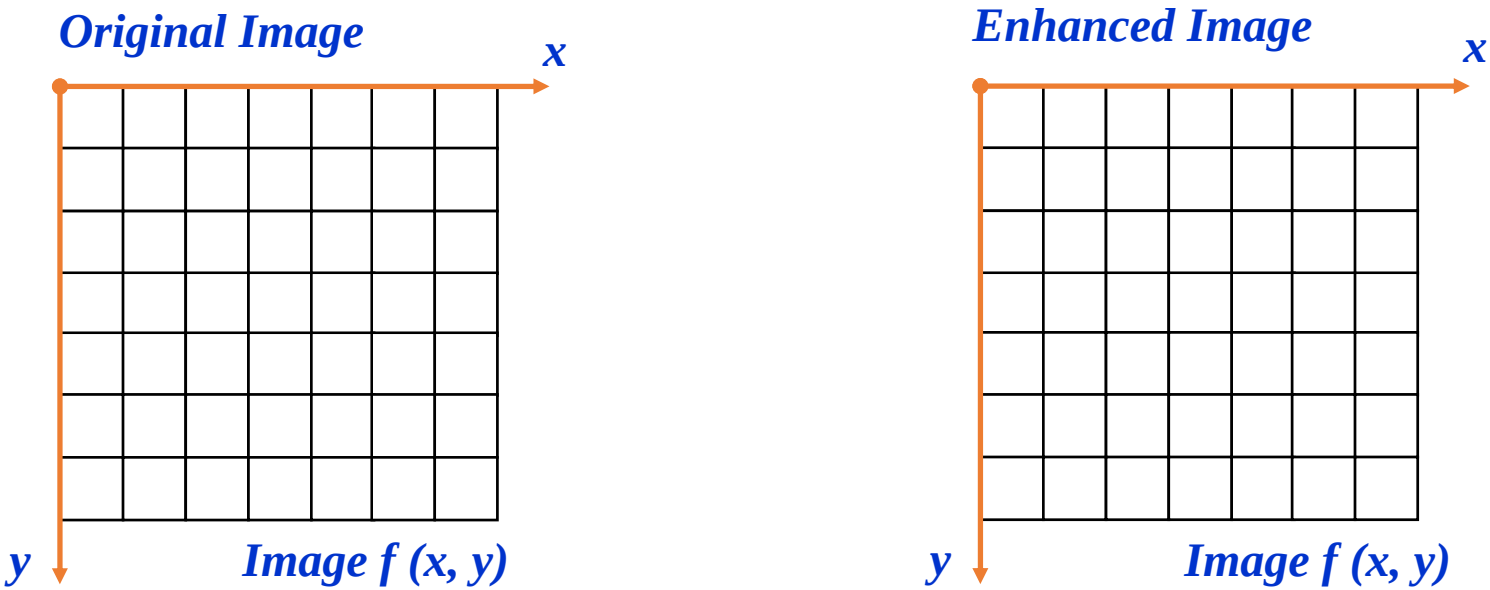
Logarithmic Transformations (cont…)

- Log functions are particularly useful when the input grey level values may have an extremely large range of values
- In the following example the Fourier transform of an image is put through a log transform to reveal more detail



Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Logarithmic Transformations (cont…)



$$s = log(1 + r)$$

We usually set c to 1

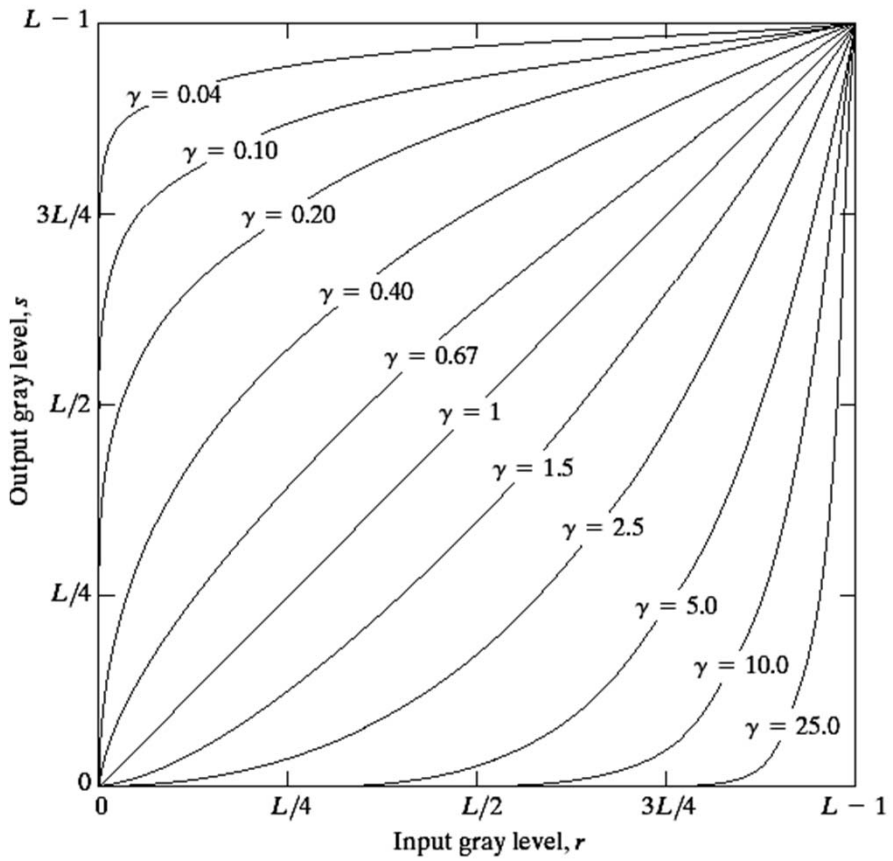
Grey levels must be in the range [0.0, 1.0]

Power Law Transformations

- Power law transformations have the following form
- $s = c * r^{\gamma}$

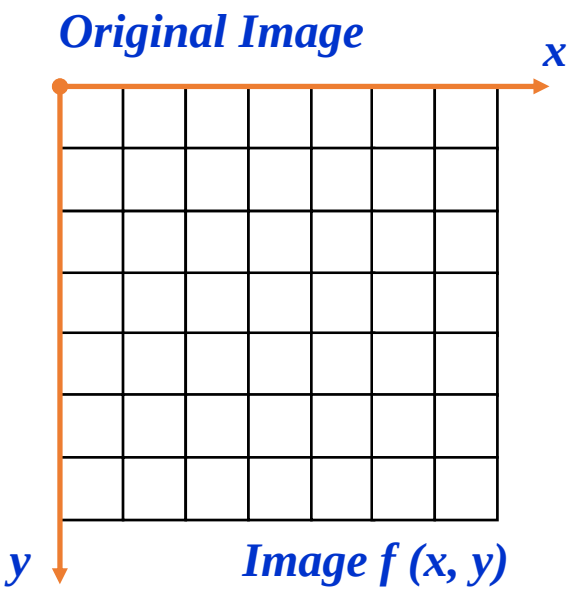
- Map a narrow range of dark input values into a wider range of output values or vice versa

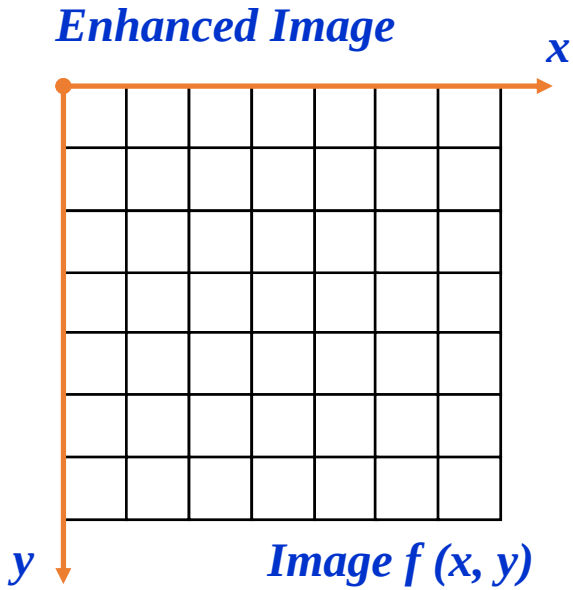
- Varying γ gives a whole family of curves



Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Power Law Transformations (cont…)





$$s = r^{\gamma}$$

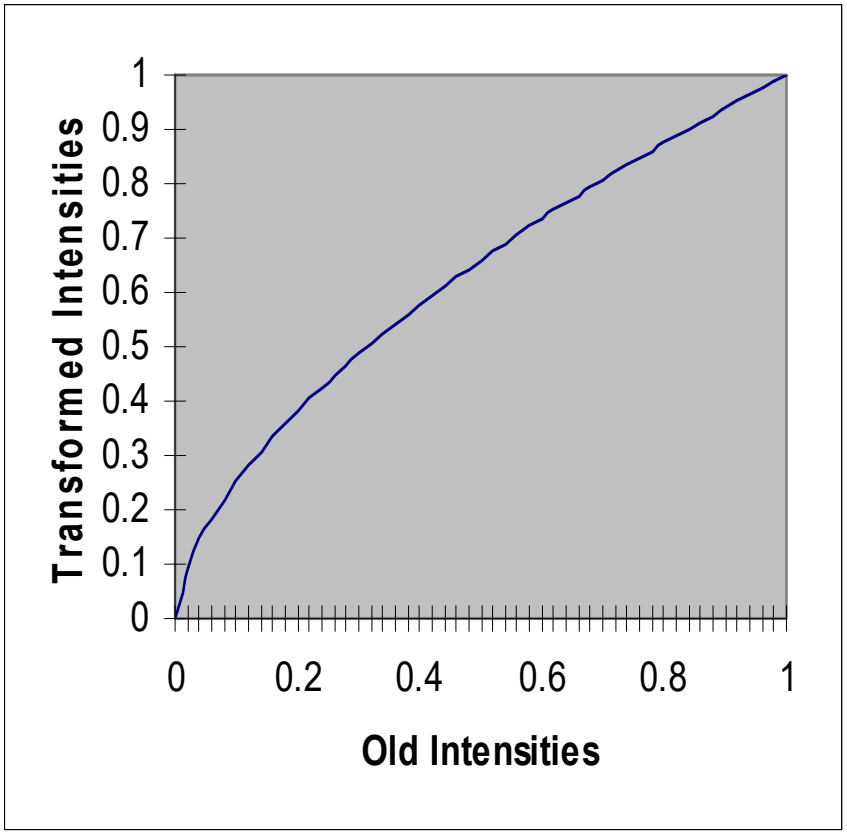
- We usually set C to 1
- Grey levels must be in the range [0.0, 1.0]

Power Law Example



Power Law Example (cont…)

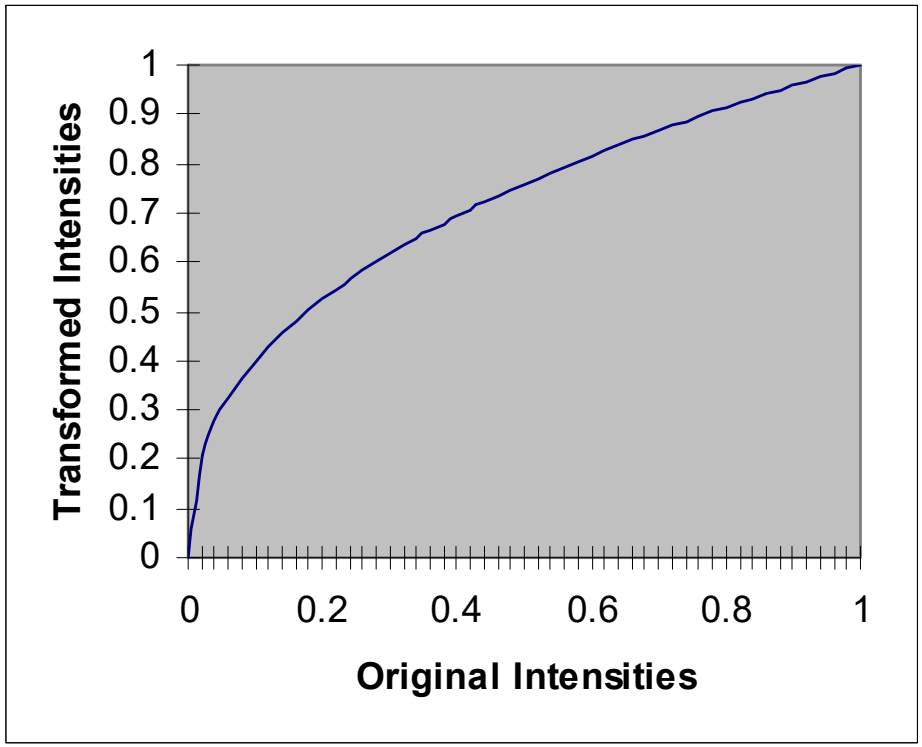
$$\gamma = 0.6$$





Power Law Example (cont…)

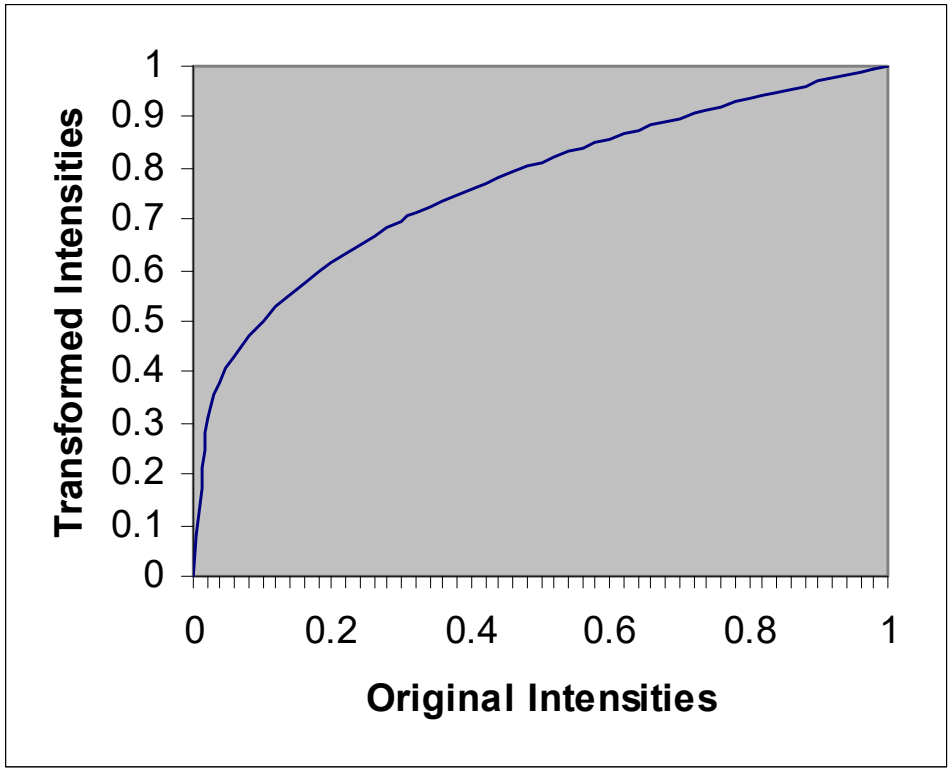
$$\gamma = 0.4$$





Power Law Example (cont…)

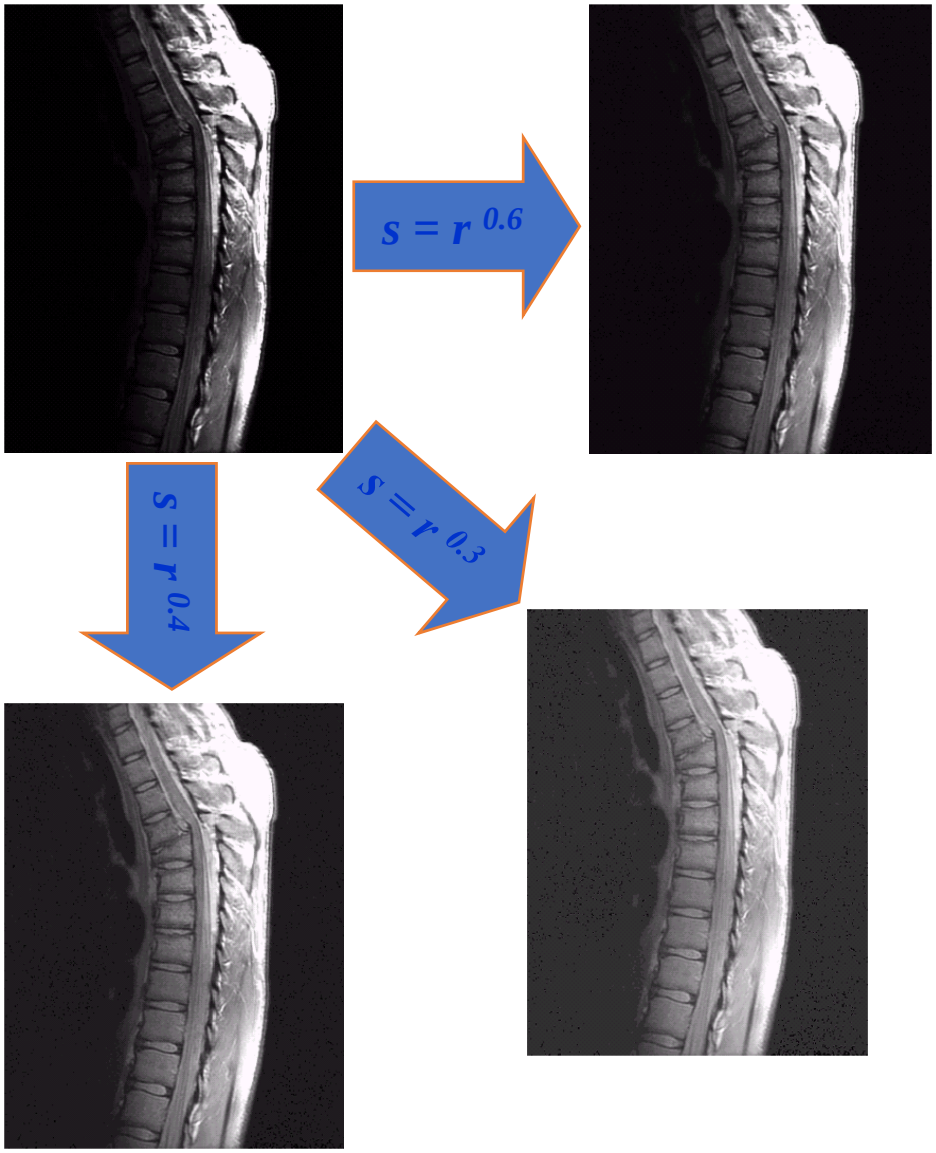
$$\gamma = 0.3$$





Power Law Example (cont…)

- The images to the right show a magnetic resonance (MR) image of a fractured human spine
- Different curves highlight different detail



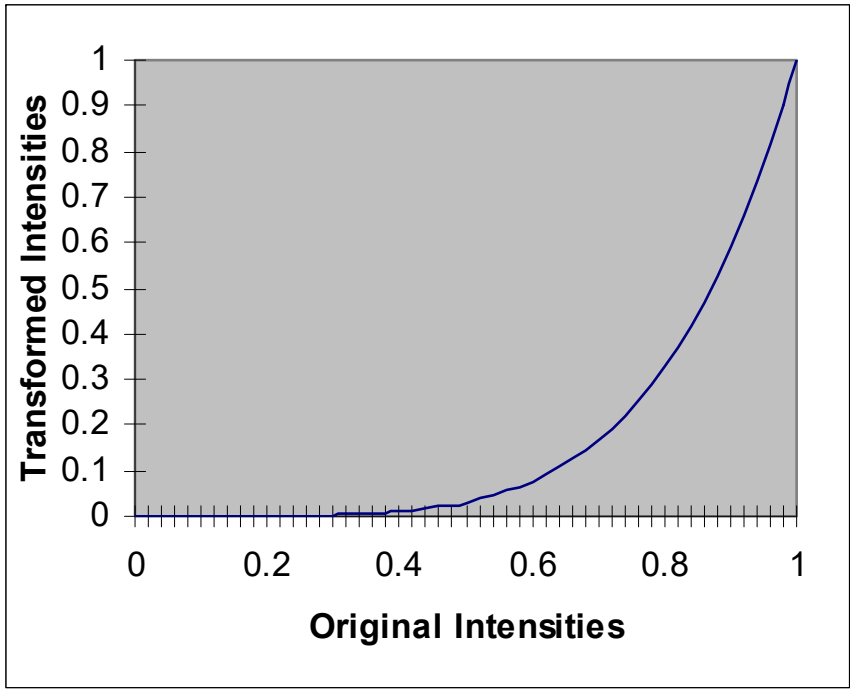
Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Power Law Example



Power Law Example (cont…)

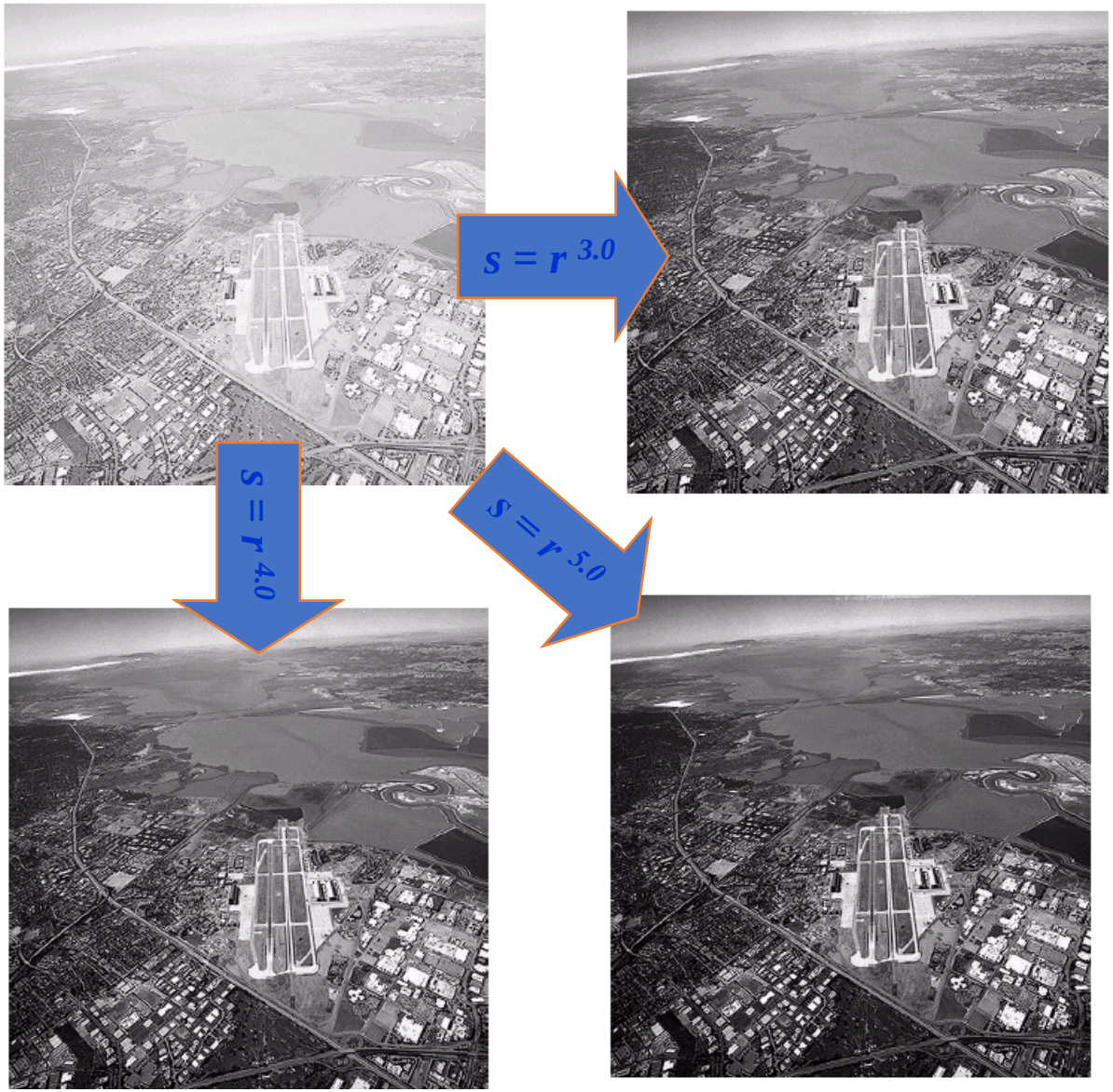
$$\gamma = 5.0$$





Power Law Transformations (cont…)

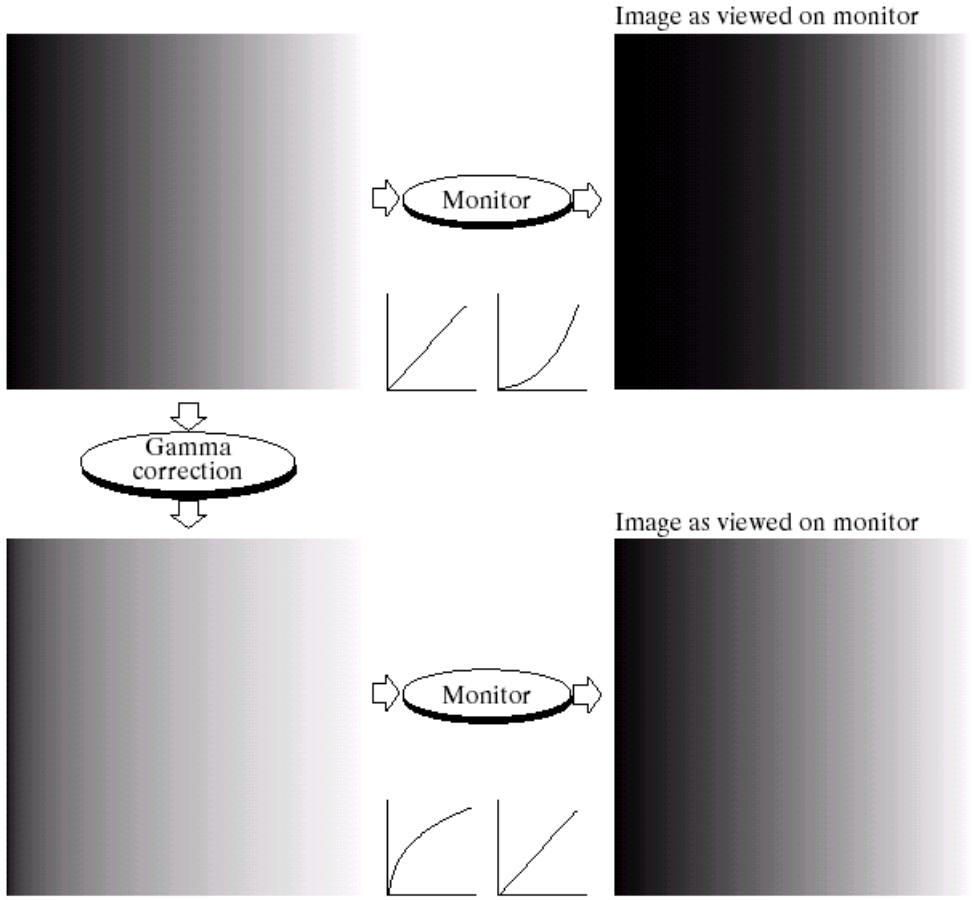
- An aerial photo of a runway is shown
- This time power law transforms are used to darken the image
- Different curves highlight different detail



Images taken from Gonzalez & Woods, Digital Image Processing (2002)

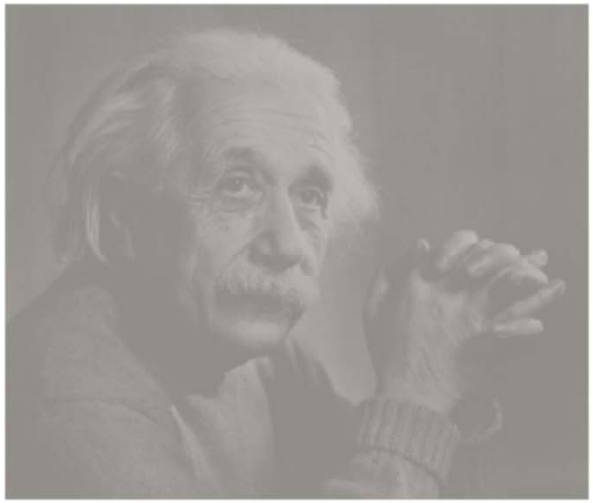
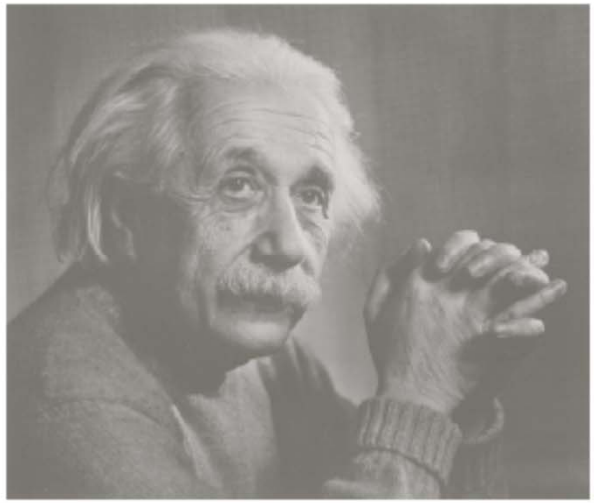
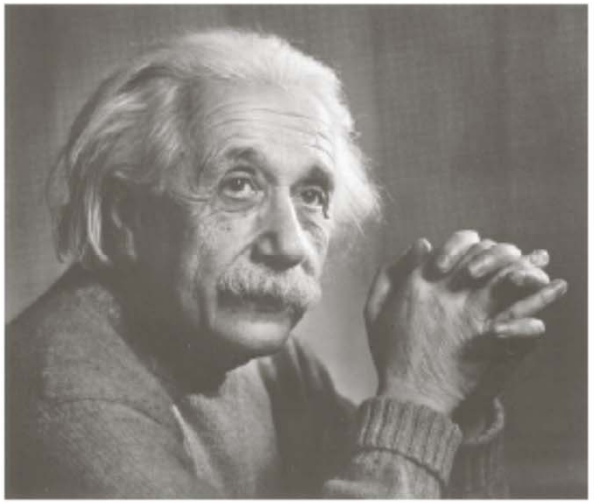
Gamma Correction

- Many of you might be familiar with gamma correction of computer monitors
- Problem is that display devices do not respond linearly to different intensities
- Can be corrected using a log transform



Images taken from Gonzalez & Woods, Digital Image Processing (2002)

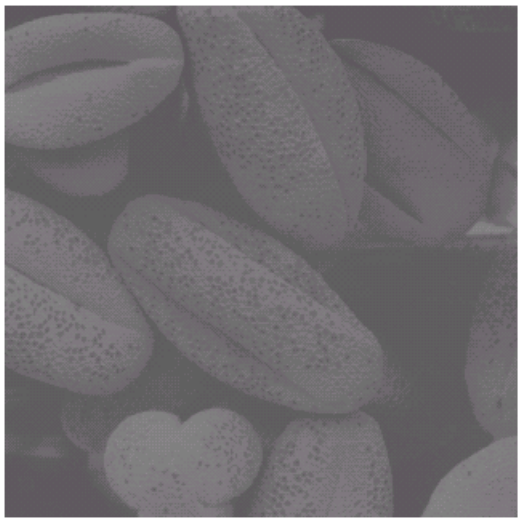
More Contrast Issues

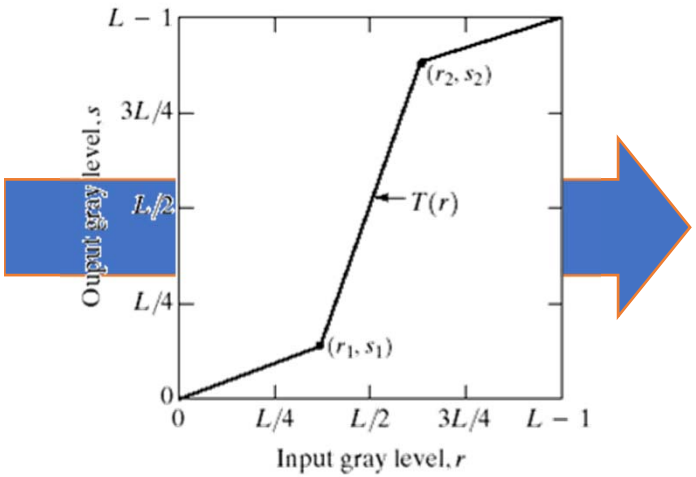




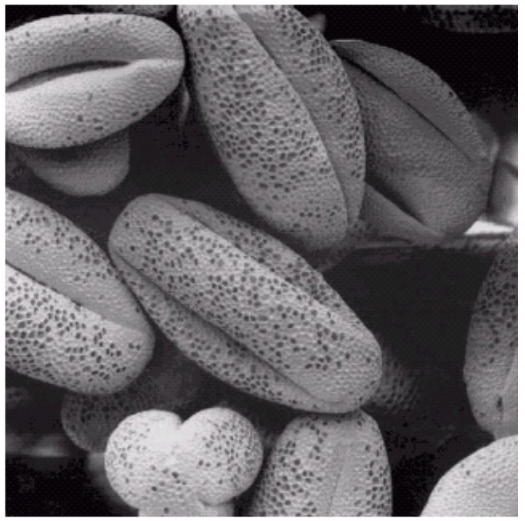
Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Piecewise Linear Transformation Functions

- Rather than using a well defined mathematical function we can use arbitrary user-defined transforms
- The images below show a contrast stretching linear transform to add contrast to a poor quality image



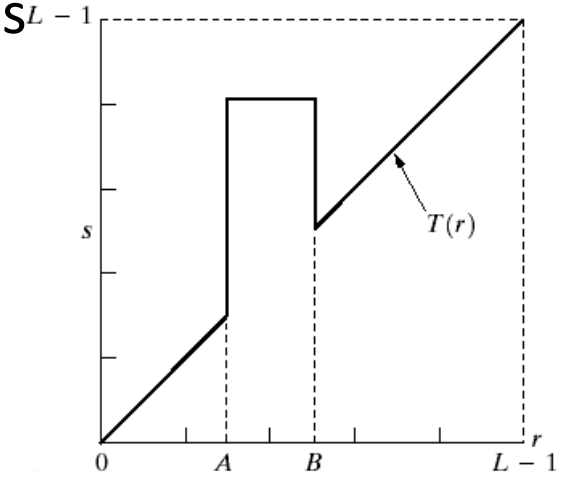


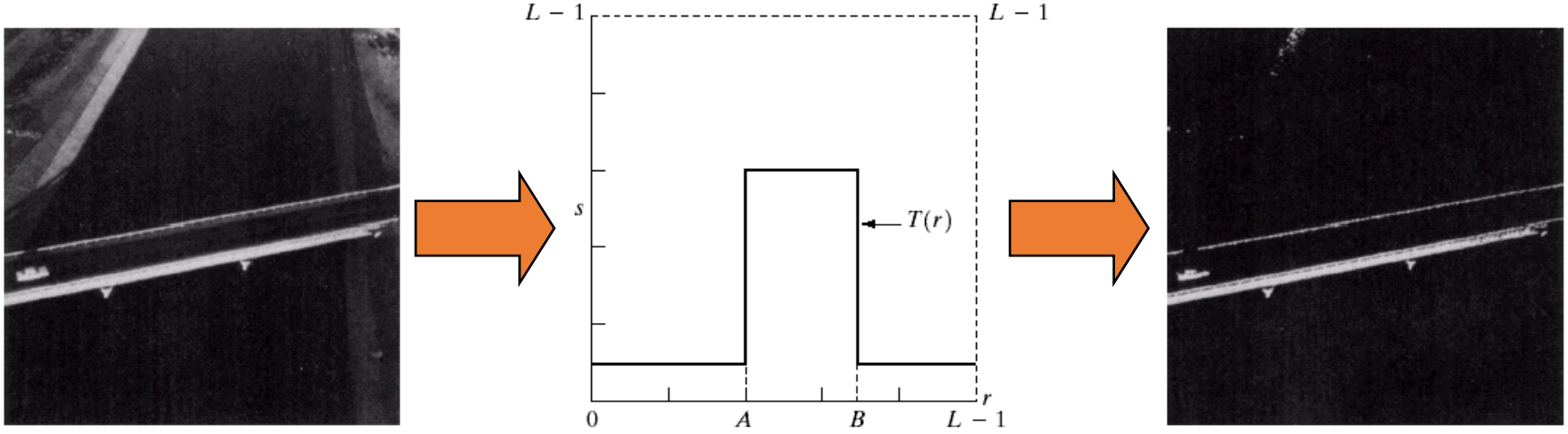


Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Gray Level Slicing

- Highlights a specific range of grey levels
  - Similar to thresholding
  - Other levels can be suppressed or maintained
  - Useful for highlighting features in an image

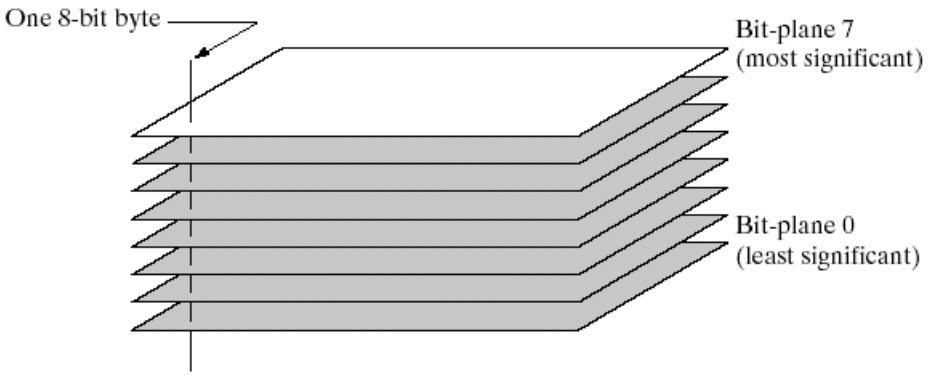


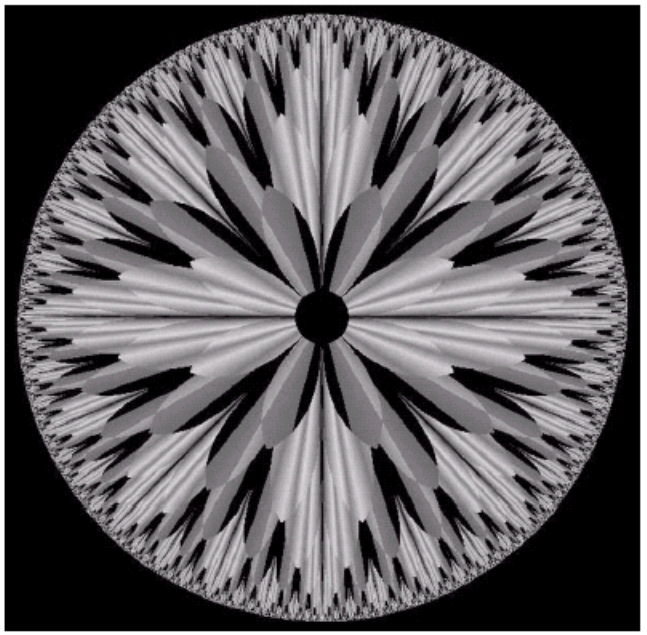


Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Bit Plane Slicing

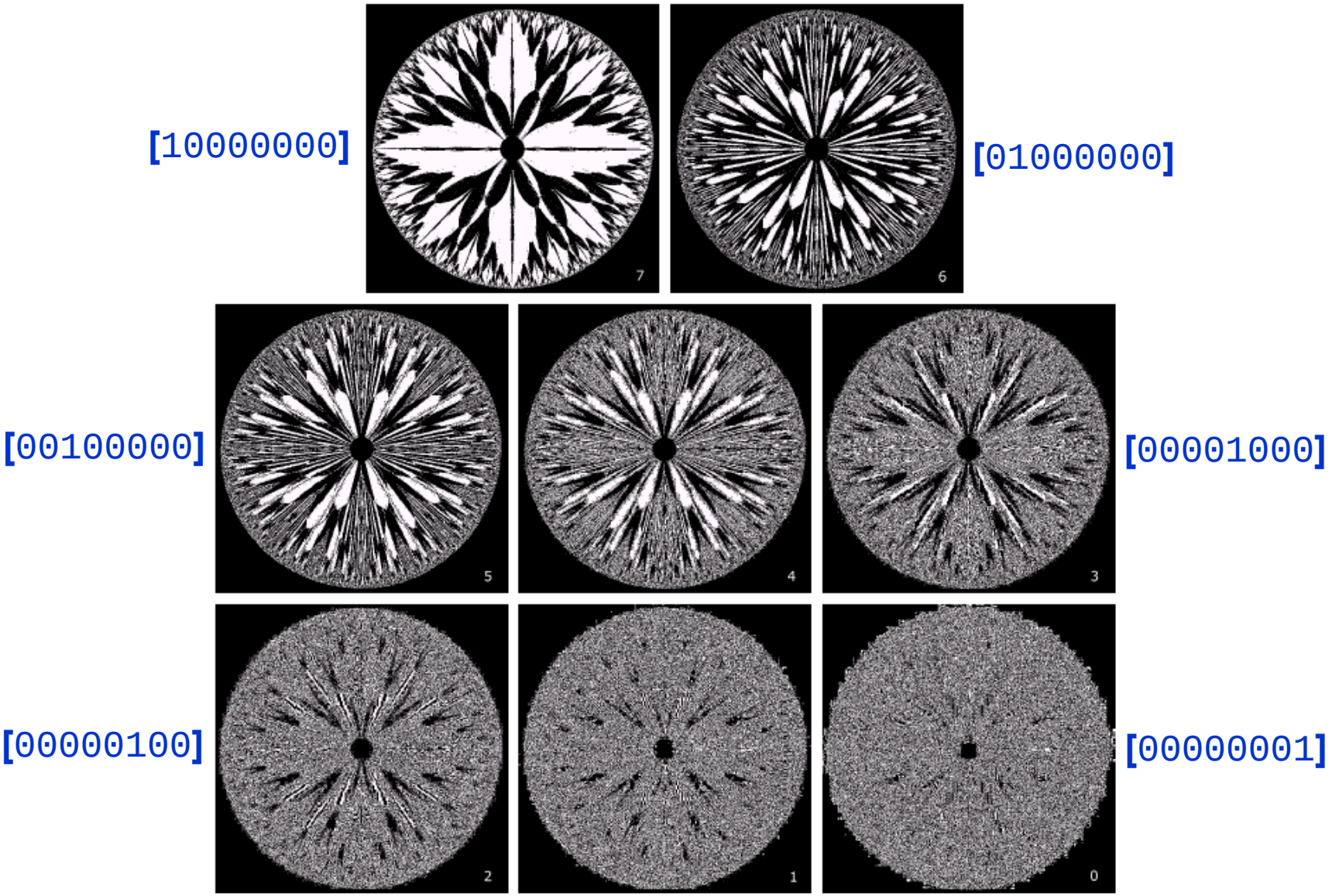
- Often by isolating particular bits of the pixel values in an image we can highlight interesting aspects of that image
  - Higher-order bits usually contain most of the significant visual information
  - Lower-order bits contain subtle details





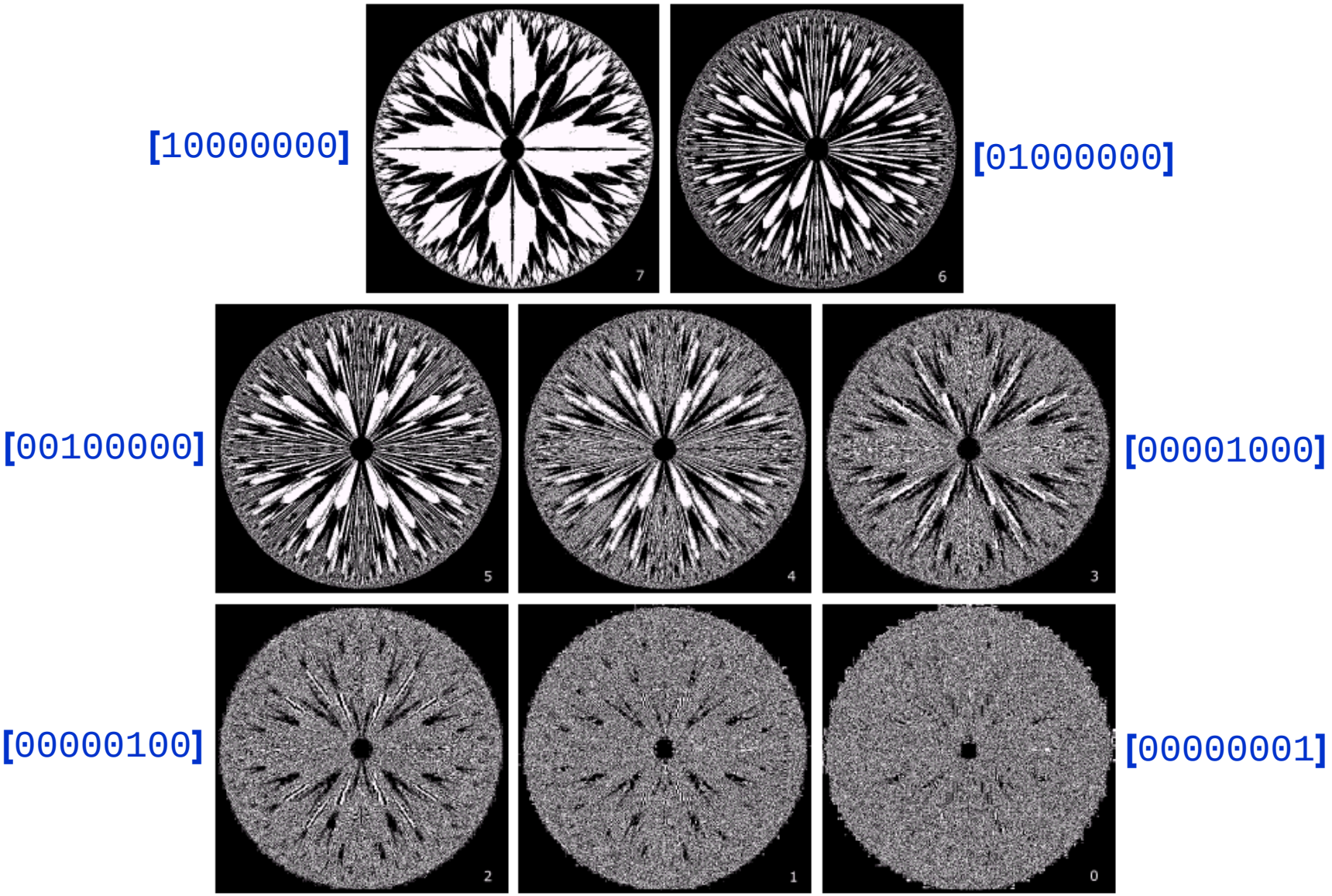
Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Bit Plane Slicing (cont…)



Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Bit Plane Slicing (cont…)



Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Bit Plane Slicing (cont…)



a b c
d e f
g h i

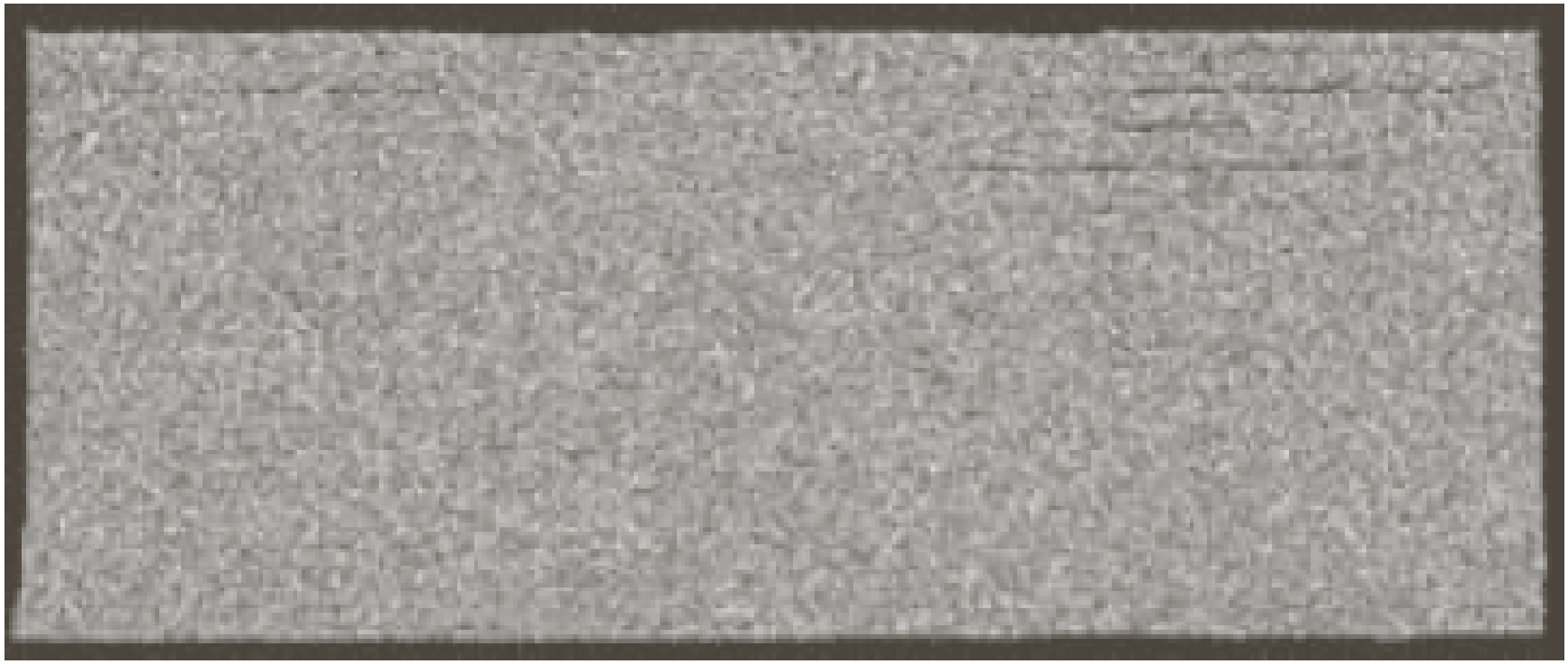
FIGURE 3.14 (a) An 8-bit gray-scale image of size 500 × 1192 pixels. (b) through (i) Bit planes 1 through 8, with bit plane 1 corresponding to the least significant bit. Each bit plane is a binary image.

Bit Plane Slicing (cont…)



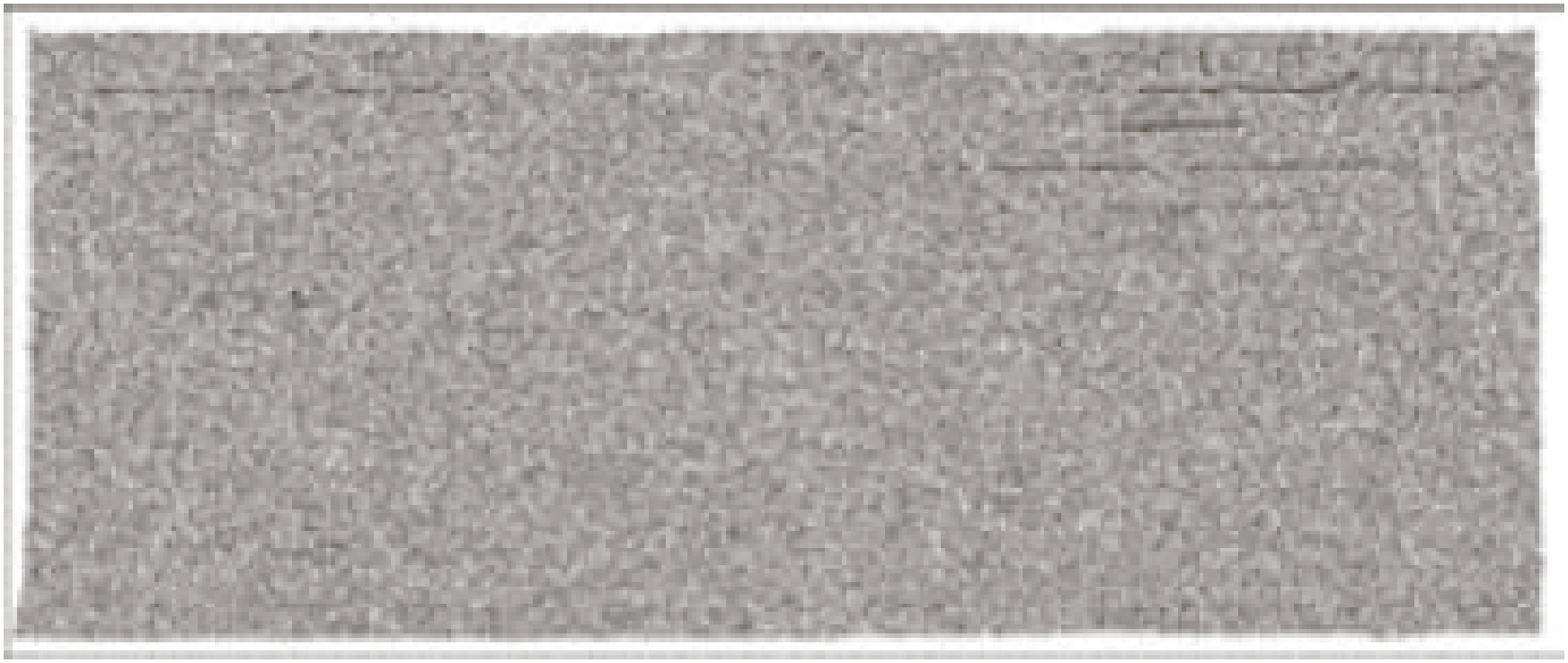
Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Bit Plane Slicing (cont…)



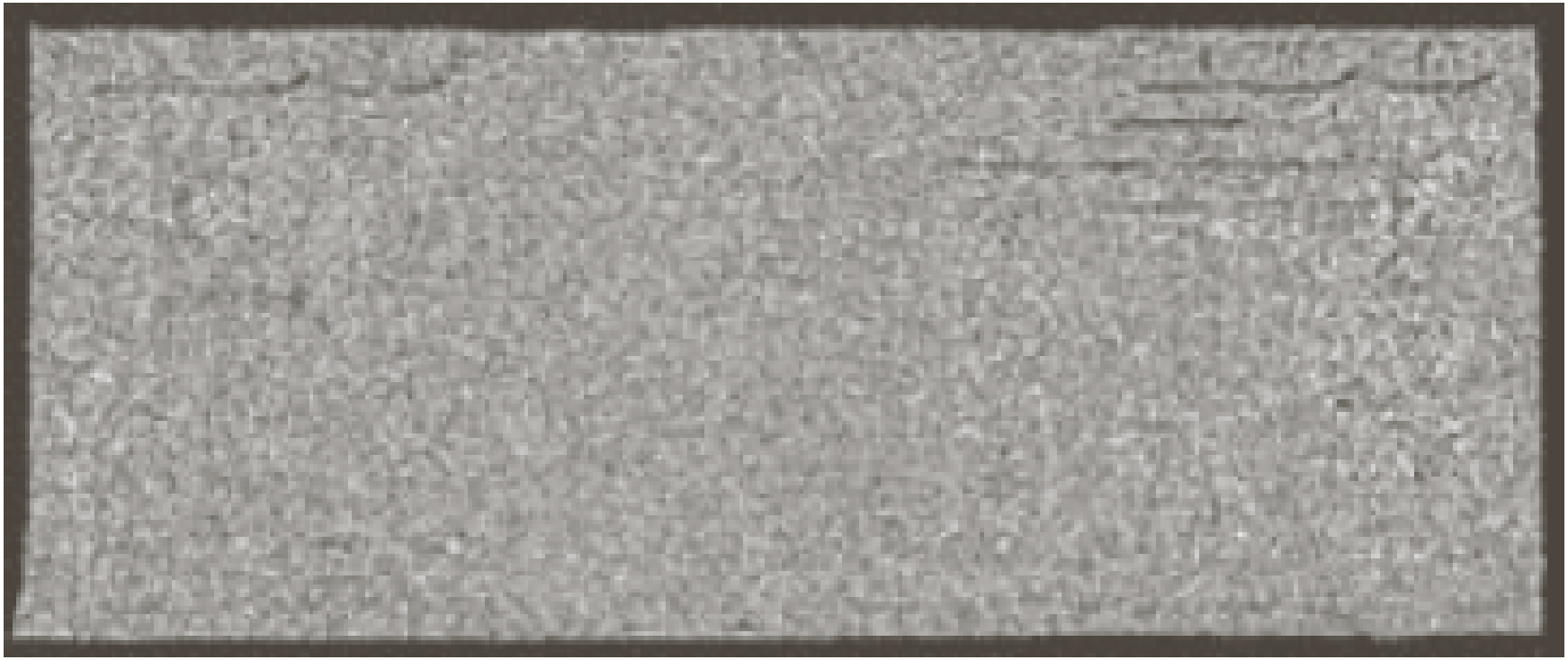
Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Bit Plane Slicing (cont…)



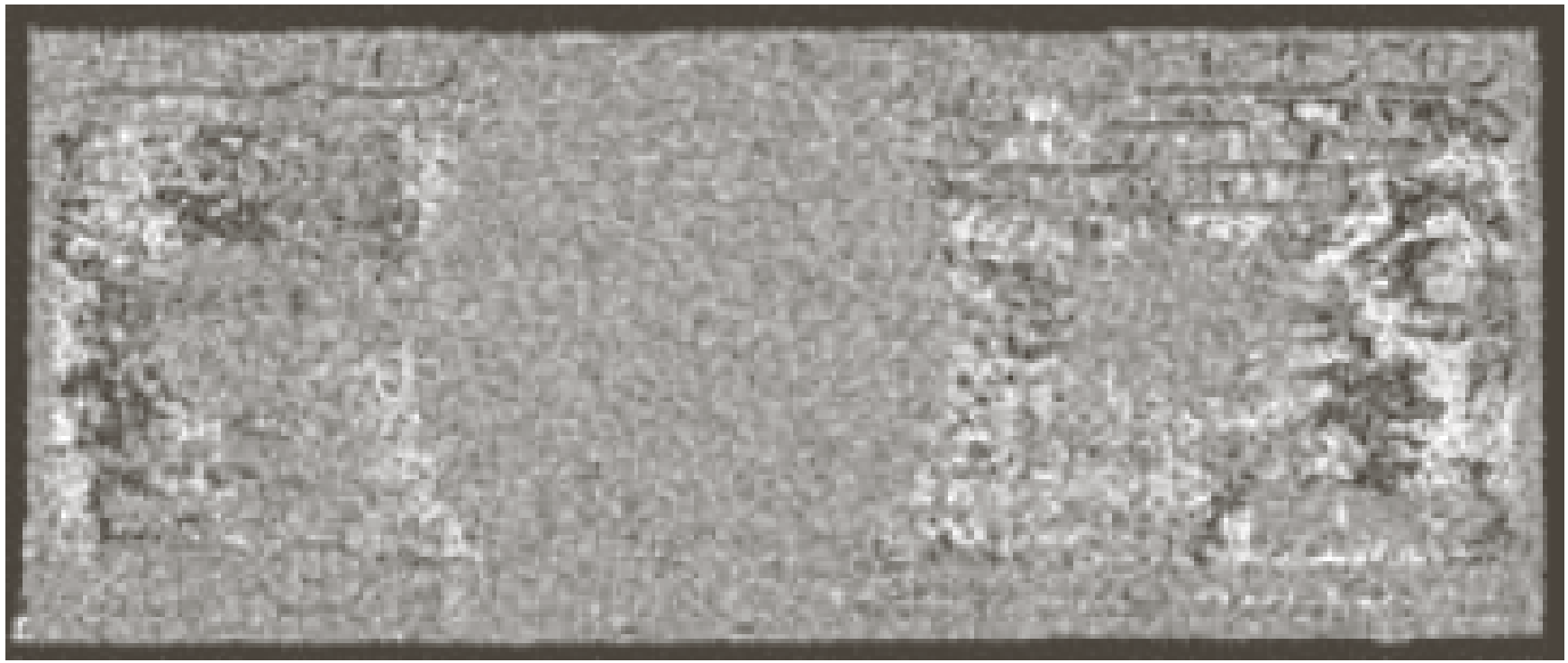
Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Bit Plane Slicing (cont…)



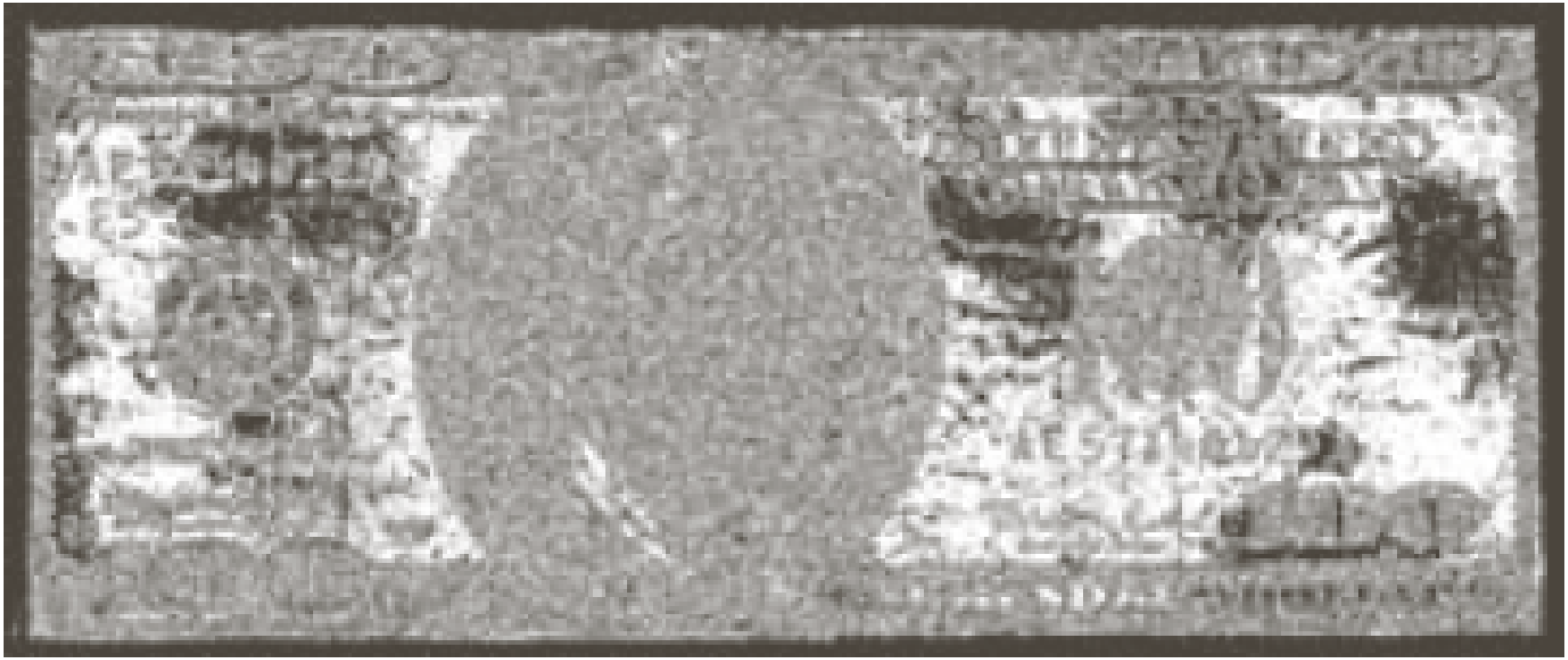
Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Bit Plane Slicing (cont…)



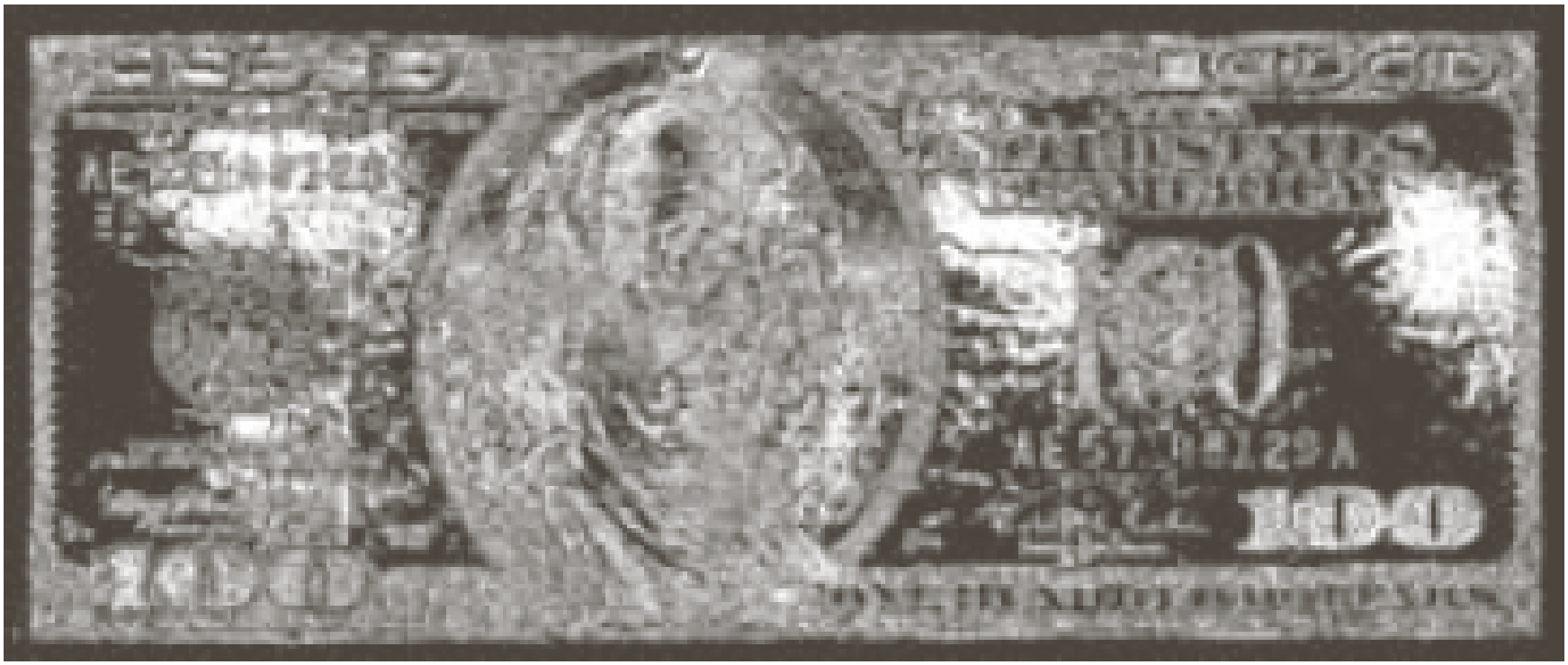
Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Bit Plane Slicing (cont…)



Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Bit Plane Slicing (cont…)



Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Bit Plane Slicing (cont…)



Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Bit Plane Slicing (cont…)



Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Bit Plane Slicing (cont…)



Reconstructed image using only bit planes 8 and 7



Reconstructed image using only bit planes 8, 7 and 6



Reconstructed image using only bit planes 7, 6 and 5

Images taken from Gonzalez & Woods, Digital Image Processing (2002)

Summary

- We have looked at different kinds of point processing image enhancement
- Next time we will start to look at neighbourhood operations – in particular *filtering* and *convolution*